

Datasets

The reference vision dataset and its manifest

EdgePilot AI - Team 15
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2 documents in this volume

datasets/ - benchmark input data

Work package: vision benchmark.

What this folder is

The controlled data a benchmark is run against. One dataset lives here today, for the vision workload.

What happened here

Not restructured. What changed is what the code around it will now accept: the vision module used to require exactly this dataset's seven labels, and a dataset with a different number of classes would run to completion and then fail validation. That restriction was removed, so this folder is now the project's *reference* dataset rather than its only possible one - a user can supply their own from the browser, and it never touches the server.

Files

Subfolder only. See `vision-benchmark/`.

Connected folders

- `datasets/vision-benchmark/` - the dataset itself.
- `src/modules/vision-benchmark/` - the code that loads and measures against it.
- `scripts/vision-benchmark/` - the generator that produces the images and the validator that checks them.
- `docs/vision-benchmark/` - the dataset card and workload specification.

datasets/vision-benchmark/ - the reference vision dataset

Work package: vision benchmark.

What this folder is

A synthetic construction-safety dataset: 21 samples across seven labels - hardhat, safety_vest, gloves, goggles, mask, ladder, safety_cone.

Synthetic on purpose. The images are generated from a committed script rather than collected, which means the dataset has no licensing question, no privacy question, and no person in it - and it can be regenerated byte-identically on any machine. `manifest.json` records a SHA-256 so a run can prove which version of the dataset it measured.

What happened here

The dataset is unchanged. The measurement taken against it is new, and it is worth recording here because this is where a reader will come looking for it: `llava:latest` scored **66.7% accuracy** on these 21 samples - 14 correct - and **failed** the workload's own 0.80 accuracy gate. The per-class matrix closes exactly: all three hardhat samples were classified as `safety_cone`, giving hardhat an F1 of 0.0 while `safety_cone` picked up three false positives. Two gloves and two masks were also called hardhats.

The `images/` subfolder is generated and git-ignored. `npm run vision:fixtures` rebuilds it, and both `prebuild` and `pretest` call that automatically, so it is never a missing prerequisite.

Files

File	What it is
<code>manifest.json</code>	The dataset contract: schema version, dataset and workload ids, manifest version and SHA-256, title, description, source, licence, the seven labels, privacy checks, generation parameters, and the 21 sample records
<code>README.md</code>	Human description of the dataset and how to regenerate it
<code>images/</code>	Generated PNGs. Git-ignored; rebuilt by <code>npm run vision:fixtures</code>

Connected folders

- `scripts/vision-benchmark/` - `generate-fixtures.mjs` writes `images/`; `validate-dataset.ts` checks this manifest against them.
- `src/modules/vision-benchmark/infrastructure/` - `manifest-loader.ts` reads this file.
- `docs/vision-benchmark/` - the dataset card that documents provenance and licence.
- `evidence/vision-benchmark/` - the measurements taken against it.

Referenced from

These folders point here. Each link below resolves in both directions.

- datasets/
- src/components/vision/
- src/modules/vision-benchmark/
- tests/vision-benchmark/